

Extra challenges: building tables

Level 1

Exercise 1

Create a table suppliers with supplier_id (primary key), name (must have a value and must be unique), country (Sweden if nothing is given) and email.

Expected: Runs without errors

Exercise 2

Add a supplier without giving a country. Then show all suppliers. What does the country column say?

Expected: 1 row, country is Sweden

Exercise 3

Try to add a second supplier with the same name, Nordic Textiles. What happens, and which rule stopped you?

Expected: An error

Exercise 4

Create a table coupons where the code itself (for example 'SUMMER20') is the primary key. discount_percent must be between 1 and 90, and valid_until must have a value. Then try to add a coupon with 95 percent off.

Expected: The table is created, the INSERT gives an error

Level 2

Use the SQLite documentation: sqlite.org/lang_createtable.html and sqlite.org/lang_altertable.html

Exercise 5

Add a supplier without giving a supplier_id. Which id does it get? Why?

Look up: INTEGER PRIMARY KEY · **Expected:** 1 new row

Exercise 6

Rename the column email in suppliers to contact_email.

Look up: ALTER TABLE ... RENAME COLUMN · **Expected:** Runs without errors

Exercise 7

Show the columns of the products table using SQL, not the Database Structure tab.

Look up: PRAGMA table_info · **Expected:** 5 rows, one per column

Exercise 8

A product can come from many suppliers, and a supplier can deliver many products. Create the middle table product_suppliers with a purchase price that must be more than 0. The same pair can only appear once. Then try to connect product 1 to supplier 99.

Look up: PRIMARY KEY with two columns · **Expected:** The table is created, the INSERT gives an error

Exercise 9

Create a table `campaigns` with `name`, `start_date` and `end_date`. Add a rule that `end_date` can never be before `start_date`. Test it with a campaign that ends before it starts.

Look up: `CHECK` that compares two columns · **Expected:** The table is created, the `INSERT` gives an error

Level 3

Exercise 10

Create a table `product_sizes` with an own `id`, `product_id` (points to products), `size` (only S, M, L or XL) and `stock` (0 if nothing is given). The same product can't have the same size twice. Test it by adding size M for product 1 twice.

Look up: `UNIQUE` on two columns · **Expected:** The first `INSERT` works, the second gives an error

Exercise 11

Create a table `employees` where each employee can have a manager, who is also an employee in the same table. The boss has no manager. Add the boss and two employees who report to the boss.

Look up: a foreign key that points to its own table · **Expected:** 3 rows

Exercise 12

Create two tables: `teams` and `players`, where a player belongs to a team. Make it so that when a team is deleted, its players are deleted automatically. Add one team with two players, delete the team, and check the `players` table.

Look up: `ON DELETE CASCADE` · **Expected:** 0 rows left in `players`